Answer key: Lesson 8, Times Stories

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Mei has 3 bags. Each bag has 5 sweets. Write the number in each box of the strip. Then fill in the blank.

Answer: 5, 5, 5 in the boxes; 15 sweets

There are 3 bags, so 3 boxes, and each box is 5 sweets. 5 + 5 + 5 = 15, so Mei has 15 sweets in all.

Watch for: Writing 3 in the boxes.; Adding 3 + 5 = 8.

S2. Ben has 4 boxes. Each box has 3 pencils. How many pencils does Ben have in all?

Answer: 4 groups of 3 = 12 (3 groups of 4 = 12 is also right); 12 pencils

4 boxes with 3 pencils in each box: 3 + 3 + 3 + 3 = 12. So 4 groups of 3 = 12 and Ben has 12 pencils. Either order gives the right total: if a child writes 3 groups of 4 = 12, ask 'how many boxes?'.

Watch for: Writing 4 + 3 = 7 by adding.; Drawing 3 boxes with 4 in each box (the total is right, so ask 'how many boxes?').

S3. Siti has 5 bags. Each bag has 2 sweets. Which strip shows Siti's sweets? Write its number in the bracket.

Answer: Strip 2

Strip 2 has 5 boxes with 2 in each box: 5 bags of 2 sweets. Strip 1 has 2 boxes of 5. Strip 3 has 5 boxes but the last box has 1, so the boxes are not equal.

Watch for: Choosing strip 1 because it has the same numbers.; Not noticing that the last box of strip 3 is smaller.

S4. Ali has 3 boxes with 5 pens in each box. He also has 2 loose pens. How many pens does Ali have in all?

Answer: 15 pens in the boxes; 17 pens in all

Step 1: 3 boxes of 5 pens is 5 + 5 + 5 = 15. Step 2: 15 and 2 loose pens make 17 pens in all.

Watch for: Writing 5 + 3 + 2 = 10.; Forgetting the loose pens.

S5. Mei has 2 bags. Each bag has 6 sweets. Draw a strip. Then fill in the blank.

Answer: A strip of 2 boxes with 6 in each box; 12 sweets

2 bags with 6 sweets in each bag: draw 2 equal boxes and write 6 in each. 6 + 6 = 12, so Mei has 12 sweets in all.

Watch for: Drawing 6 boxes with 2 in each.; Drawing boxes that are not the same size.

S6. Ravi has 4 bags with 3 marbles in each bag. He loses 2 marbles. How many marbles does Ravi have now?

Answer: 12 marbles in the bags; 10 marbles now

Step 1: 4 bags of 3 marbles is 3 + 3 + 3 + 3 = 12. Step 2: he loses 2, so 12 - 2 = 10 marbles are left.

Watch for: Adding the 2 marbles.; Taking 2 away from 4 or from 3.

Think harder (page 3)

T1. Jenny has 3 boxes with 6 pens in each box and 5 loose pens. She gives 8 pens away.

Answer: 18 pens; 23 pens; 15 pens left

Step 1: 3 boxes of 6 is 6 + 6 + 6 = 18. Step 2: 18 and 5 loose pens make 23. Step 3: 23 take away 8 is 15 pens left.

Watch for: Stopping at 23.; Taking away 8 from 18 before adding the loose pens.

T2. Mei has 7 marbles. Ali has twice as many marbles as Mei.

Answer: 14 marbles; 21 marbles together

Mei's strip is 7. Ali has twice as many: 7 and 7, which is 14. Together they have 7 + 14 = 21 marbles.

Watch for: Writing 9 (7 + 2).; Writing 14 for together.

T3. Mei has 4 stickers. Ravi has twice as many stickers as Mei. Jenny says Ravi has 6 stickers.

Answer: No; 8 stickers

Mei has 4, so Ravi has two strips of 4: 4 + 4 = 8 stickers. Jenny added 2 to 4 to get 6, but twice as many is not 2 more. So Jenny is not correct.

Watch for: Saying Yes because 4 and 2 make 6.; Writing 4 or 12.

T4. Ali has twice as many cards as Mei. Ali has 24 cards. Ali gives some cards to Mei. Then they have the same number of cards.

Answer: Mei 12 cards; Ali gives 6 cards

Ali has two strips of 12, so Mei has 12. The gap is 12. Ali gives half of the gap: two equal parts of 12 are 6 and 6, so he gives 6. Then Ali has 24 - 6 = 18 and Mei has 12 + 6 = 18. They have the same number.

Watch for: Giving all 12 cards.; Writing 12 for (b).

Go further (page 4)

G1. Wei puts 3 stickers on each of 4 cards. He puts 2 stickers on each of 5 other cards. He also has 4 stickers in his hand.

Answer: 12 stickers; 10 stickers; 26 stickers at first

4 cards with 3 stickers: 3 + 3 + 3 + 3 = 12. 5 cards with 2 stickers: 2 + 2 + 2 + 2 + 2 = 10. 12 + 10 + 4 = 26 stickers at first (the 4 in his hand count too).

Watch for: Forgetting the 4 stickers in his hand.; Using 3 for both sets of cards.

G2. Siti has 5 marbles. Ravi has 3 times as many marbles as Siti.

Answer: 15 marbles; 20 marbles altogether

Siti's strip is 5. Ravi has 3 times as many: 5 + 5 + 5 = 15. Together: 5 + 15 = 20 marbles.

Watch for: Writing 8 (5 + 3).; Writing 15 for altogether.

G3. Kumar has 3 times as many pens as Lily. They have 24 pens together.

Answer: Lily 6 pens; Kumar 18 pens

Lily has 1 box and Kumar has 3 boxes, so there are 4 equal boxes. 24 pens in 4 equal boxes is 6 in each box (6 + 6 + 6 + 6 = 24). Lily has 6 and Kumar has 18. Check: 3 × 6 = 18 and 6 + 18 = 24.

Watch for: Writing 8 (24 shared into 3).; Writing 3 times 24.

Investigation (page 5): Twice, and twice again

Mei 3, Ali 6, Ben 12, Jenny 24. Each child has twice as many as the child before, so each strip is twice as long as the one before. The numbers double each time: 3, 6, 12, 24.

Look for: A child who draws or builds each strip as two copies of the strip before, writes 3, 6, 12, 24 and notices that each number is double the last. Credit a child who says the strips get twice as long, or who checks 12 + 12. Do not mark down a child who finds only Ali and Ben.

Exit ticket (page 5)

E1. Ali has 4 bags. Each bag has 5 sweets. Which number sentence finds how many sweets Ali has in all? Write its number in the bracket.

Answer: (2) 4 × 5

4 bags with 5 sweets in each bag is 4 groups of 5, which is 4 × 5. Adding 4 + 5 joins the numbers of bags and sweets. 5 - 4 finds a gap.

Watch for: Choosing 4 + 5.; Choosing 5 - 4.

E2. Lily has 2 bags with 5 marbles in each bag. She also has 3 loose marbles. How many marbles does Lily have in all?

Answer: 13

Step 1: 2 bags of 5 marbles is 10. Step 2: 10 and 3 loose marbles make 13 marbles in all.

Watch for: Writing 10 (forgetting the loose marbles).; Writing 2 + 5 + 3 = 10.

E3. Ravi has 3 bags with 4 sweets in each bag. He eats 2 sweets. How many sweets does Ravi have now?

Answer: 10

Step 1: 3 bags of 4 sweets is 12. Step 2: he eats 2, so 12 - 2 = 10 sweets are left.

Watch for: Adding the 2 sweets: 14.; Writing 12 (forgetting the 2 he eats).

Sneak-peek puzzles (not in the workbook)

1. 12
2. 15. Ben has twice as many as Mei: 5 and 5, which is 10. Mei has 5 and Ben has 10, so together they have 15.